



**Title: Comprehensive Analysis of Healthcare,  
Employment, and Educational Data**

# Introduction & Overview

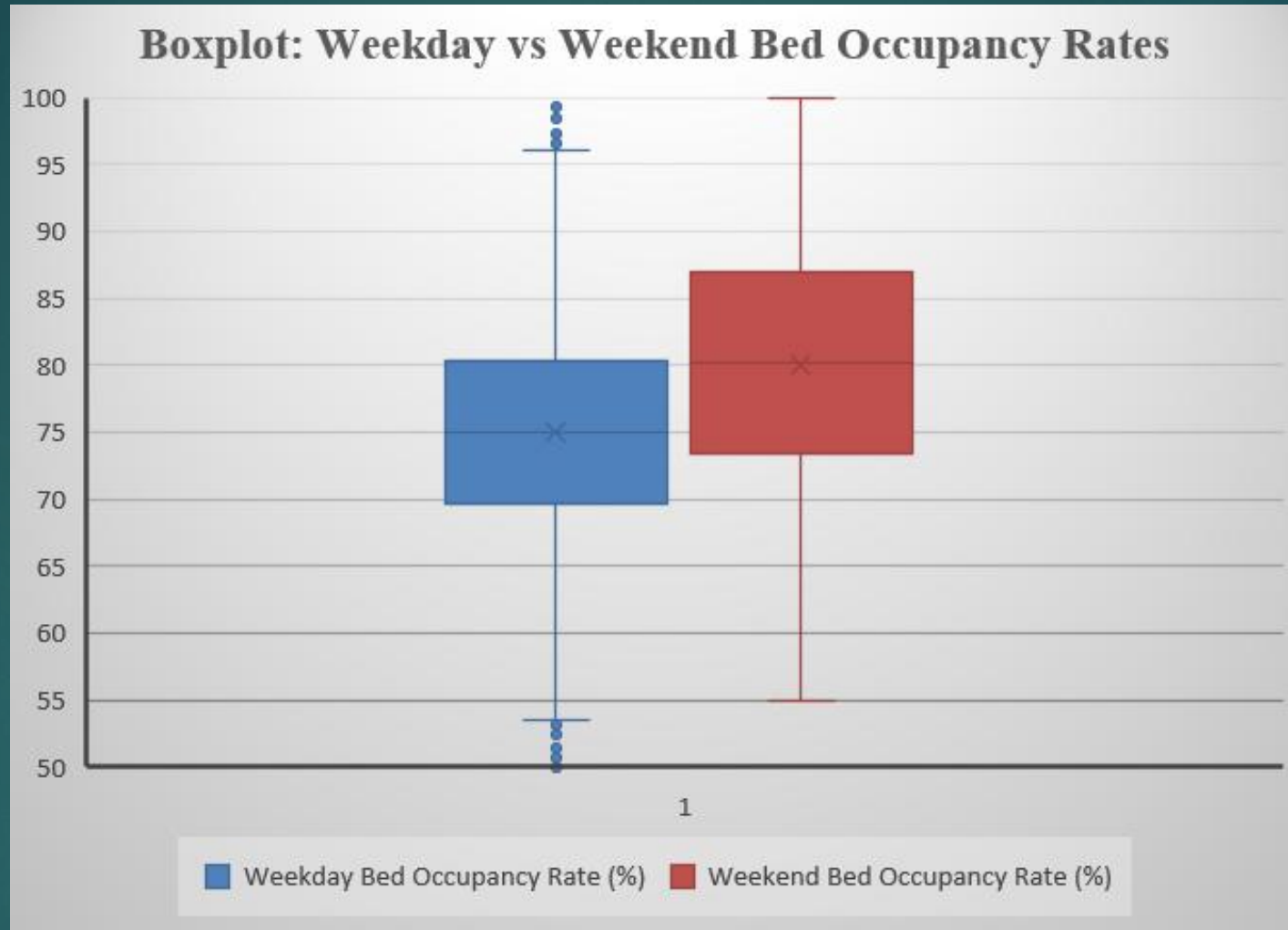
## Data Visualization Analysis - Four Key Studies

- **Hospital Bed Occupancy:** Weekday vs Weekend patterns (5,840 records)
- **Global Graduate Salaries:** 111 graduates across 5 majors & regions
- **Clinical Trial Bias:** Weight-loss study with 249 participants
- **Economic Correlation:** Unemployment vs Loan delinquency (27 US cities)
- ▶ **Objective:** Apply data visualization techniques to extract actionable insights across healthcare, education, and economics

# Hospital Bed Occupancy - Key Findings

**Title: Weekends Show Higher Hospital Occupancy**

- ▶ **Key Results:**
- ▶ • **Weekday Average:** 74.97%  $\pm$  8.05%
- ▶ • **Weekend Average:** 80.11%  $\pm$  9.76%
- ▶ • **Difference:** 5.14 percentage points higher on weekends
- ▶ • **Statistical Significance:**  $p < 0.001$  (highly significant)
- ▶ **Implications:** • Hospitals need 5% extra capacity on weekends • Higher variability requires flexible staffing plans



**Figure 1.** Boxplot comparing weekday and weekend hospital bed occupancy rates (%). Outliers are shown as circles.

# Global Graduate Salaries - Major & Regional Analysis

**Title: Tech Fields Pay Best, Location Matters**

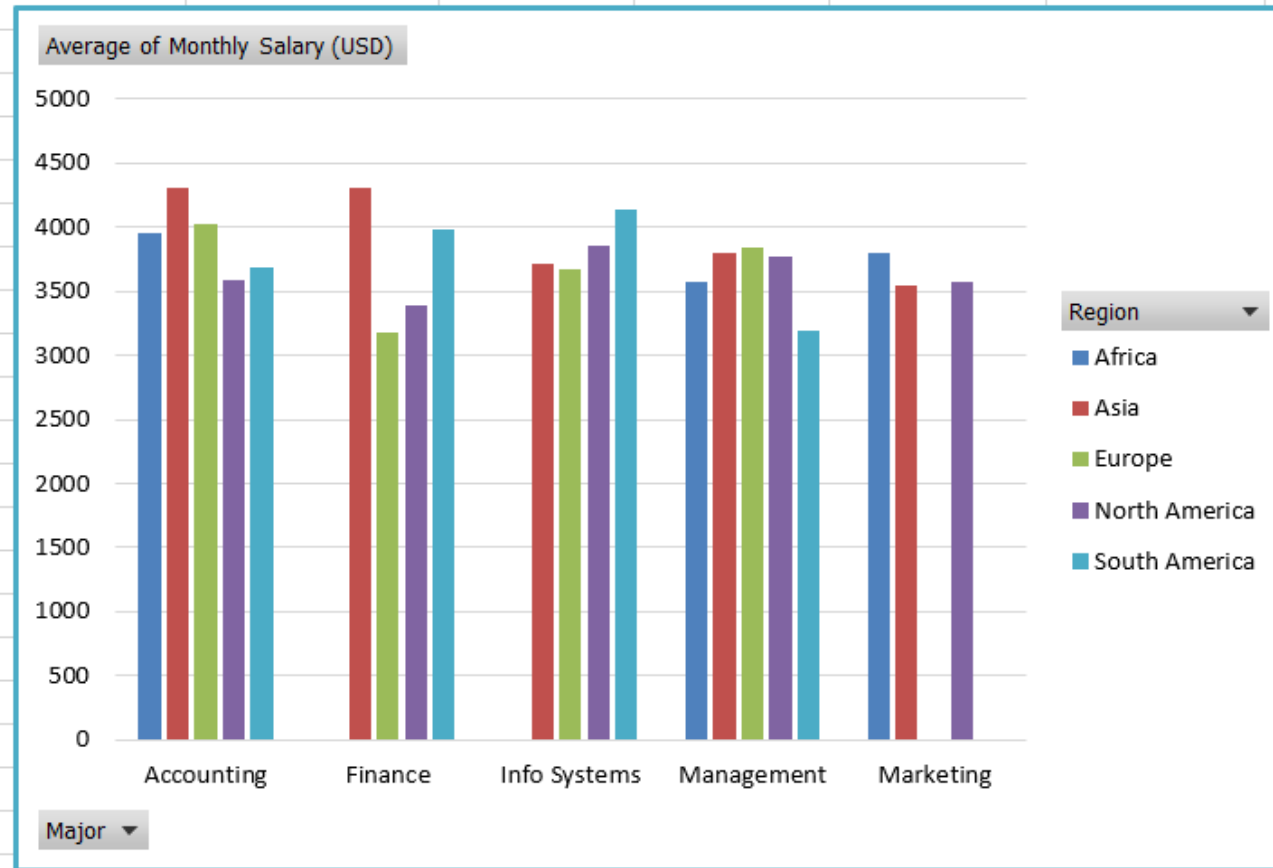
## Top Earning Majors:

- ▶ **Information Systems** - Highest overall salaries
- ▶ **Finance** - Strong earning potential
- ▶ **Management** - Moderate levels
- ▶ **Marketing** - Moderate levels
- ▶ **Accounting** - Lowest across regions

## Regional Rankings:

- ▶ • **Highest:** North America & Europe
- ▶ • **Lowest:** Africa & South America

| Average of Monthly Salary (USD) | Column Labels  |                    |             |                    |                 |                    |
|---------------------------------|----------------|--------------------|-------------|--------------------|-----------------|--------------------|
| Row Labels                      | Africa         | Asia               | Europe      | North America      | South America   | Grand Total        |
| Accounting                      | 3947.5         | 4299.75            | 4022.142857 | 3585.1             | 3685.666667     | 3870.0625          |
| Finance                         |                | 4308               | 3179        | 3393               | 3980.333333     | 3623.823529        |
| Info Systems                    |                | 3716.8             | 3666        | 3854.571429        | 4133            | 3793.5625          |
| Management                      | 3577           | 3796.75            | 3834.222222 | 3776.4             | 3188            | 3773.323529        |
| Marketing                       | 3804           | 3551.666667        |             | 3568.333333        |                 | 3618.916667        |
| <b>Grand Total</b>              | <b>3880.75</b> | <b>3877.740741</b> | <b>3689</b> | <b>3693.763158</b> | <b>3789.875</b> | <b>3764.540541</b> |



**Figure 2.** Average Starting Salaries by Major Across Regions

# Salary Distribution Analysis

## Title: Career Progression & Salary Patterns

### Experience Level Impact:

- ▶ • **Entry Level:** 2,500 – 3,500 *range*
- ▶ • **Mid Level:** 3,500-4,500 range
- ▶ • **Senior/Executive:** \$4,500-6,000 range

### Distribution Shapes:

- ▶ • **IS & Finance:** Broad, favorable spreads
- ▶ • **Management:** Bell curve, predictable
- ▶ • **Accounting:** Most standardized/concentrated

| Finding (Q1, Median, Q3) |                     |                  |                       |                    |                     |
|--------------------------|---------------------|------------------|-----------------------|--------------------|---------------------|
|                          | Management salaries | Finance salaries | Info Systems salaries | Marketing salaries | Accounting salaries |
| Q1 (25th Percentile)     | 3133.5              | 2968.5           | 3136.5                | 3026.5             | 3407.5              |
| Median (50th Percentile) | 3913                | 3709             | 3681                  | 3774               | 3807                |
| Q3 (75th Percentile)     | 4300.25             | 4327.75          | 4371.5                | 4309.5             | 4340.25             |
|                          |                     |                  |                       |                    |                     |

**Figure 3.** Salary Percentiles (25th, 50th, 75th) by Major



# Clinical Trial Bias Analysis

## Title: Three Critical Biases Affecting Results

- ▶ **Study Context:** 249 participants, 47% completion rate, 3.2 kg/m<sup>2</sup> BMI reduction
- ▶ **Three Major Biases:**
- ▶ **Selection Bias**
  - ▶ Volunteers ≠ general population
  - ▶ Results may not generalize
- ▶ **Survivor Bias ⚠ CRITICAL**
  - ▶ Only 47% completed study
  - ▶ 53% dropouts excluded from results
- ▶ **Simpson's Paradox**
  - ▶ Overall results may hide subgroup differences
  - ▶ Age, gender, BMI variations masked
- ▶ **Conclusion:** True effectiveness likely much lower than reported 3.2 kg/m<sup>2</sup>

# Unemployment vs Delinquent Loans

## Title: Weaker Relationship Than Expected

### ▶ Statistical Results:

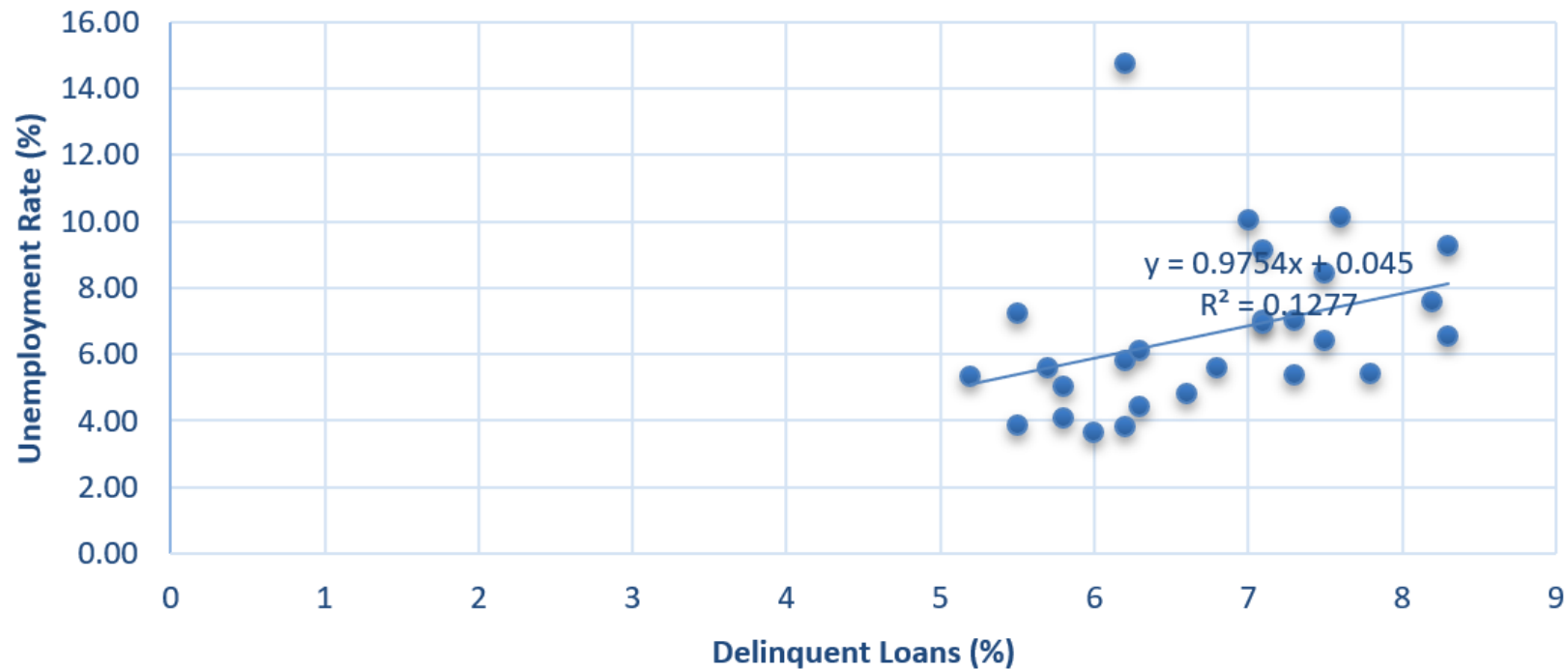
- ▶ • **Correlation (r):** 0.357 (weak positive)
- ▶ • **R<sup>2</sup> Value:** 0.1277 (only 12.77% explained)
- ▶ • **Significance:**  $p = 0.067$  (not significant at 5% level)



### ▶ Key Findings:

- ▶ • **Outlier:** Miami shows unusually high delinquency
- ▶ • **Trend:** Linear but very weak
- ▶ • **Implication:** Other factors more important than unemployment alone

## Relationship between Unemployment Rate and Delinquent Loans (%)



**Figure 4.** Scatter plot of Unemployment Rate (%) vs. Delinquent Loans (%) with linear trendline ( $R^2 = 0.1277$ ).

# Key Insights & Practical Applications

## ▶ **Healthcare Management:**

- ▶ • Weekend capacity planning crucial (5% higher demand)
- ▶ • Variable demand requires flexible resource allocation

## ▶ **Career Planning:**

- ▶ • Choose **Information Systems** or **Finance** for highest salaries
- ▶ • **North America/Europe** offer best opportunities
  - Experience level significantly impacts earnings

## ▶ **Research Quality:**

- ▶ • **Survivor bias** can severely overestimate treatment effects
- ▶ • Always analyze dropouts and subgroups

## ▶ **Economic Policy:**

- ▶ • Unemployment alone poor predictor of loan problems
- ▶ • Need comprehensive economic indicators

# Conclusions

## ▶ Key Takeaways:

- ✓ **Statistical significance** doesn't always mean practical importance
- ✓ **Visualization** reveals patterns not obvious in raw data
- ✓ **Bias analysis** essential for valid research conclusions
- ✓ **Geographic and field differences** drive major salary variations

## ▶ Data Visualization Success:

- ▶ Box plots revealed distribution differences
- ▶ Scatter plots showed weak correlations  
Histograms displayed salary distribution shapes
- ▶ Statistical analysis confirmed significance